patenting in biotechnology ensures fair financial returns for those who have invested finances, ideas, time and hard work for an invention.

The following criteria must be satisfied for patenting :

- The invention must be novel and useful;
- The product must be inventive and reproducible;
- The patent application should provide the full description of the invention and the invention must be patentable.



The first living organism that was patented was a genetically engineered species of bacteria - *Pseudomonas putida* in 1980 which was genetically engineered by Ananda Mohan Chakrabarty in 1971.

General agreement of tariffs and trade (GATT) and trade related IPRs (TRIPs)

GATT was framed in 1948 by developed countries to settle dispute, among the countries regarding share of world trade. The benefits of GATT was enjoyed only by developed countries. In 1988 US congress enacted a law 'the omnibus trade and competitiveness act' (OTCA) which gave powers to US to investigate the laws related to trade.

Geographical indication (GI)

A geographical indication is a name or sign used on products which correspond to a specific geographical origin and possess qualities or a reputation that are due to that origin. Geographical indications are typically used for agricultural products, food products, handicrafts and industrial products. Darjeeling tea was the first GI tagged product in India in 2004-05. In Tamilnadu, Kancheepuram silk, Coimbatore wetgrinder, Thanjavur paintings, Madurai Malli, Thirunelveli Halwa and Temple jewellery of Nagercoil are GI tagged.

Copyright

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Trade marks

Any specific symbol or words to identify a particular product or process of a company constitute trademark. This enables the public to distinguish between a trader's goods from similar goods of other traders

Biotechnology as an independent discipline has drawn world wide attention from the Governments and the corporate world because of its limitless applications. It is looked upon as a panacea for treating diseases and genetic disorders. The global demand of the biotechnological products is on the increase. It is the science for the future with solutions to many of the problems related to health, agriculture, environment and industries.

Summary

Biotechnology is defined as "any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use". In 1919, Hungarian agricultural engineer Karl Ereky coined the term Biotechnology. Biotechnology includes two major technologies, Genetic engineering and Chemical engineering.

Biotechnology has application in four major industrial areas, including health care (medical) agriculture, industrial and environment.

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